

Low-level Visual Reasoning on Vector Graphics											
	Tools	AC	LC	SW-S 2Obj	SW-S mObj	SW Sup	NLVR	Geo	Maze 2×2	Maze 3×3	All
Monolithic Large Multimodal Models											
Llava-1.5-7b	-	0.53	0.49	0.48	0.55	0.35	0.53	0.50	0.00	0.00	0.381
Llava-1.5-13b	-	0.53	0.51	0.51	0.47	0.61	0.48	0.50	0.00	0.00	0.401
Gllava-7b	-	0.59	0.50	0.43	0.54	0.43	0.49	0.58	0.00	0.00	0.396
GPT-4V	-	0.58	0.64	0.77	0.60	0.61	0.63	0.64	0.28	0.02	0.530
GPT-4o	-	0.63	0.57	0.97	0.82	0.92	0.81	0.71	0.46	0.08	0.663
Visual Programming with LLM (text-only) reasoner											
ViperGPT (w/ GPT-4)	CI	0.11	0.67	0.61	0.47	0.53	0.43	0.02	0.03	0.00	0.319
VDLM with LLM (text-only) reasoners											
VDLM-txt (w/ GPT-4)	-	0.89	0.95	0.78	0.63	0.80	0.68	0.63	0.40	0.19	0.661
VDLM-txt (w/ GPT-4)	CI	0.73	0.95	0.89	0.68	0.72	0.72	0.64	0.40	0.26	0.666
VDLM with LMM (multimodal) reasoners											
VDLM-mm (w/ GPT-4V)	-	0.55	0.94	0.84	0.62	0.72	0.71	0.69	0.60	0.20	0.652
VDLM-mm (w/ GPT-4o)	-	0.90	0.95	0.91	0.82	0.82	0.86	0.71	0.61	0.34	0.769

Table 1: Zero-shot accuracy on low-level visual reasoning tasks. Task abbreviations: AC (Angle Classification), LC (Length Comparison), SW-S-2Obj/mObj (Shapeworld Spatial Reasoning with two objects or multiple objects), SW-Sup (Shapeworld Superlative), Geo (Geoclidean 2-shot Learning). “CI” refers to Code Interpreter. VDLM-mm brings consistent overall improvements to GPT-4V and GPT-4o, as indicated by the comparison within the blue and orange rows. Detailed analysis is presented in §3.3 and §4.

High-level Visual Reasoning on Vector Graphics				
VGBench-QA				
	Category	Color	Usage	All
Llava-v1.5-7b	0.26	0.32	0.27	0.283
Llava-v1.5-13b	0.32	0.43	0.39	0.380
Gllava-7b	0.16	0.33	0.21	0.233
GPT-4o	0.58	0.84	0.76	0.726
VDLM-mm (w/ GPT-4o)	0.62	0.86	0.75	0.743

Table 2: Zero-shot accuracy on high-level visual reasoning tasks. We show that VDLM-mm preserves the LMM’s capability on semantic-centric reasoning that does not require precise low-level perception.

2022) dataset requiring the model to understand a compositional geometric concept with only two reference examples. (8-9) **Maze Solving**: Solve a 2×2 or 3×3 maze, given the starting and ending positions. Among these tasks, Angle Classification, Length Comparison, and Maze Solving are newly created from scratch (See Appendix K for more details).

High-level visual reasoning tasks. Although the focus of this work is on low-level visual reasoning, we additionally include a set of high-level tasks to investigate the impact of VDLM on knowledge reasoning tasks. These tasks rarely require precise perception of the locations and measurements of the primitives. We leverage VGBench (Zou et al., 2024), a benchmark originally proposed for evaluating LLMs in understanding and generating vector graphics codes. In this work, we evaluate LMMs and VDLM-mm for question-answering based on the rasterized VGBench SVG images.

Figure 4 shows simplified input and output examples for each task. Full prompts can be found in Appendix J. To reduce the cost of evaluating proprietary models, we randomly sample a subset of 100 instances for each task. We consider a zero-shot evaluation setting for all tasks. Note that the SVG-to-PVD model in VDLM is trained purely on synthesized task-agnostic data.

3.2 Models

We compare our work with strong baselines, including both state-of-the-art monolithic large multimodal models (LMMs), i.e., LLaVA-v1.5 (Liu et al., 2023a), GLLaVA Gao et al. (2023), GPT-4V (OpenAI,

2023a)*, GPT-4o (OpenAI, 2024)[†], as well as visual programming agents, e.g., ViperGPT (Surís et al., 2023). ViperGPT employs an LLM to generate code, which can call external vision models, such as GLIP (Li et al., 2022) and BLIP2 (Li et al., 2023b), to process the image and generate the final output. Given that ViperGPT-style models successfully separate perception from reasoning, we seek to investigate whether the existing perception tools adequately recognize low-level primitives in vector graphics. For VDLM, we explore two variants, namely VDLM-mm with GPT-4V, GPT-4o and VDLM-txt with GPT-4 (text-only).[‡] We also experiment with applying weaker LMM reasoners, such as LLaVA, to VDLM-mm. We find that interpreting PVD requires a certain level of text reasoning capability, and the benefits only emerge with strong LMMs, as shown in Figure 5. However, recent strong open-source LMMs, such as Qwen-2.5-VL-72B (Bai et al., 2025), can also benefit from PVD features (see Appendix F for additional experiments). To obtain more insights in comparing with ViperGPT, we further investigate augmenting VDLM-txt with a Code Interpreter (CI). We employ the GPT-4 Assistant[§] for our experiments, designating the code interpreter as the sole tool available. We use the same set of prompts for both VDLM-txt and VDLM-mm. See details about prompt design in Appendix J.

3.3 Results

Table 1 shows the zero-shot accuracy for the evaluation tasks. We outline the key findings as follows:

VDLM significantly improves LMMs on low-level reasoning tasks, while preserving their capabilities in high-level reasoning. The fact that VDLM-txt performs competitively even with a text-only reasoner highlights the efficacy of the intermediate PVD representation for precise low-level visual perception. Without any task-specific fine-tuning, strong LMM reasoners can effectively incorporate the additional information provided PVD alongside the image input. Figure 5 further demonstrates that this benefit only emerges when the LMM has a certain level of text-reasoning ability and persists in state-of-the-art LMMs. For high-level reasoning tasks (Table 2), the improvement is more subtle, as the tasks focus on the semantics of the vector graphics, such as “what can this be used for?”, which rarely require precise location or measurements of visual elements.

QA performance on complex math problems does not necessarily reflect a faithful understanding of low-level visual concepts. We observe that G-LLaVA (Gao et al., 2023), a model demonstrating strong performance on geometric problems, such as MathVista (Lu et al., 2023), still struggles with understanding basic lines and angles, which are prerequisites for solving geometric math problems.

Existing vision-language models, such as GLIP and BLIP2, are ineffective as low-level visual preceptors. This is evidenced by the unsatisfactory performance of ViperGPT, even when equipped with a strong planner like GPT-4. On the other hand, we observe that augmenting the reasoning model in VDLM-txt with code interpreters can be particularly helpful for tasks requiring algorithmic reasoning, such as 3×3 maze solving.

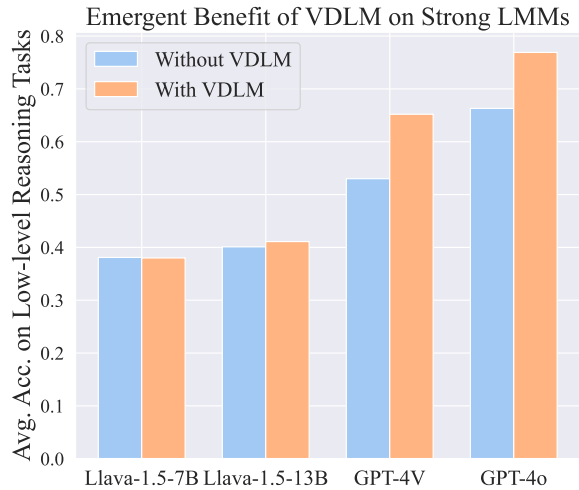


Figure 5: The direct improvements brought by VDLM to LMMs emerge when the LMM possesses sufficient text reasoning capabilities. These improvements are consistent with stronger LMMs, such as GPT-4o, which have enhanced spatial reasoning performance.

*GPT-4V model version: gpt-4-1106-vision-preview.

[†]GPT-4o model version: gpt-4o-2024-05-13

[‡]GPT-4 (text-only) model version: gpt-4-0125-preview.

[§]<https://platform.openai.com/docs/assistants/overview/agents>

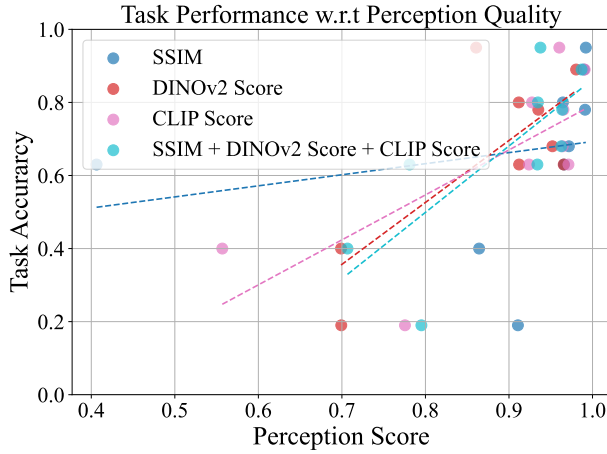


Figure 6: Task-level correlation between PVD perception quality and end-task performance.

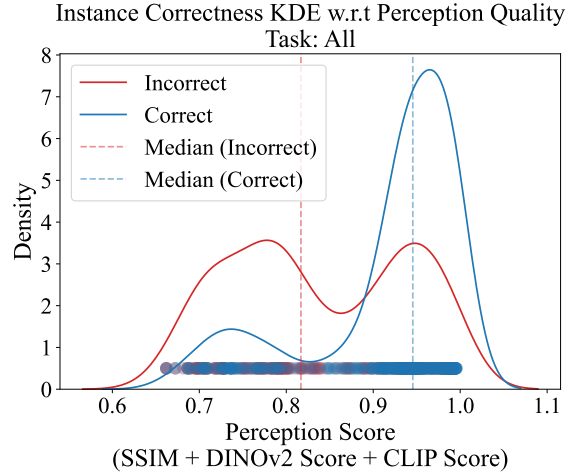


Figure 7: Instance-level correlation between PVD perception quality and end-task performance. We observe a consistent positive correlation as in task-level.

Impact of errors in PVD perception. In certain tasks, such as Shapeworld Spatial Reasoning, GPT-4o achieves better performance than VDLMM-mm. The reason lies in the imperfect perception results of the SVG-to-PVD model, despite the image-to-SVG step achieving near-perfect reconstruction as shown in Appendix D. Since the SVG-to-PVD model is trained with purely synthetic data, it is not yet perfect when generalizing to diverse domains. In Appendix E, we present an additional ablation study on the design choices of the PVD model, including alternative LLM selections and a comparison to the PNG-to-PVD approach. We also carefully analyze the remaining errors in §4.2, and demonstrate the impact of improving perception on end-task performance (§4.1). In §6, we discuss future work for developing a more general and expressive PVD representation.

4 Analysis

4.1 Primal Visual Description Quality vs End-Task Performance

One advantage of a modular system is that enhancing an individual module can lead to improvements in the overall system. In this section, we explore whether a positive correlation exists between the quality of the intermediate perception representation and end-task performance. To investigate this, we first define metrics to reflect the quality of the PVD perception. Upon generating a PVD perception result, we render it back into a raster image using our procedural image generator. We then compute a similarity score between the reconstructed image and the original input image as a measure of the perception performance. For measuring the similarity, we consider both pixel-based and embedding-based metrics. We adopt the Structural Similarity (SSIM) Index (Wang et al., 2004) score to assess pixel-level similarity. Additionally, to account for semantic similarity, we adopt a CLIP-score (Radford et al., 2021) and a DINOv2-score (Oquab et al., 2023), which are calculated as the cosine similarity of the flattened CLIP and DINOv2 embeddings, respectively.

In Figures 6 and 7, we visualize the impact of the perception quality, on the 9 low-level reasoning tasks with VDLMM-txt, at both the task and instance levels. In Figure 6, each point denotes the accuracy of a task, with different colors representing different similarity metrics. The dashed lines depict linear regression results of the points, revealing a consistent positive correlation between perception quality and task accuracy across the metrics. Since the task-level accuracy may not be directly comparable across different tasks, we additionally perform an instance-level analysis using Kernel Density Estimation (KDE) on the correctness of all task instances with respect to their perception scores. As shown in Figure 7, the “correct” distribution visibly skews to the area of higher perception scores, indicating that better perception tends to result in a correct